



Bonnyrigg High School

MATHEMATICS STANDARD 2

2023 Year 12 Course Assessment Task 1 (Reference Sheet Task)

Monday, 11 December 2023

General Instructions

- Working time – 45 minutes
- Total marks – 32 marks
- Write using black pen.
- Calculators approved by NESA may be used.
- One double-sided handwritten A4-sized reference sheet is allowed.
- For each question, show relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Sections

- I. Working with Time (Q1 – Q4)
- II. Budgeting and Household Expenses (Q5 – Q6)
- III. Rates and Ratios (Q7 – Q9)
- IV. Investments and Loans (Q10 – Q12)

STUDENT NAME:

Class: (please ✓)

☐ 12MS1H – Mr Singh

☐ 12MS3H – Mr Huang

☐ 12MS5H1 – Mrs Sharma

☐ 12MS5H2 – Mr Nguyen

☐ 12MS6H – Mr Ta

Marker's use only

QUESTION	1-4	5-6	7-9	10-12	Total	%
MARKS	$\frac{\quad}{8}$	$\frac{\quad}{8}$	$\frac{\quad}{8}$	$\frac{\quad}{8}$	$\frac{\quad}{32}$	

**Each mark is worth 3.125%*

Marker's Comments:

Section I – Working with Time

8 marks

Attempt Questions 1–4

Allow about 10 minutes for this section.

Answer the questions in the spaces provided. Your responses should include relevant mathematical reasoning and/or calculations. Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Question 1 (1 mark)

Marks

What is the time difference between 6:25 am and 8:45 pm? Both times are in the same day.

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Question 2 (2 marks)

Consider the train timetable below:

Monday to Friday	♿	♿	♿	♿	♿	♿	♿	♿	♿
Liverpool	07:51	-	-	08:04	-	08:16	-	08:32	-
Warwick Farm	07:54	-	-	08:07	-	08:19	-	08:35	-
Cabramatta	07:57	-	-	08:10	-	08:22	-	08:38	-
Carramar	08:00	-	-	-	-	08:25	-	08:41	-
Villawood	08:03	-	-	-	-	08:28	-	08:44	-
Leightonfield	08:05	-	-	-	-	08:30	-	08:46	-
Chester Hill	-	-	-	08:16	-	08:32	-	08:48	-
Sefton	-	-	-	08:18	-	08:34	-	08:50	-
Lidcombe	-	-	08:03	-	08:18	-	08:34	-	08:49
Berala	-	-	08:06	-	08:21	-	08:37	-	08:52
Regents Park	-	-	08:09	-	08:24	-	08:39	-	08:54
Birrong	-	-	08:13	08:22	08:28	08:38	08:43	08:54	08:58
Yagoona	-	-	08:16	-	08:30	-	08:45	-	09:00
Bankstown	08:15	-	08:19	08:27	08:37	08:44	08:52	09:00	09:07

(a) How long is the **first** train ride from Lidcombe to Bankstown?

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(b) Yoshua needs to arrive in Bankstown by 8:40 am or he will miss his class.

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What is the latest train he can take from Cabramatta so he can arrive at Bankstown on time?

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Question 3 (1 mark)

Marko lives in Belgrade, Serbia located at (45° N, 20° E).

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Marko wants to call Tijana, who lives in New York, USA located at (40° N, 75° W).

What is the difference in **longitude** between Belgrade and New York?

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Question 4 (4 marks)

Natalie departs from Sydney (UTC +10:00) at 11:00 am on Monday 11th December.

She flies for 12 hours and land in Hawaii, stopping for 1 hour before continuing.

She flies for a further 5 hours and 30 minutes, arriving in Los Angeles (UTC –8:00).

(a) What is the time zone difference between Sydney and Los Angeles?

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(b) Hence, determine what the **local time and day** is when Natalie lands in Los Angeles.

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Continue to Section II – Budgeting and Household Expenses

Section II – Budgeting and Household Expenses

8 marks

Attempt Questions 5–6

Allow about 10 minutes for this section.

Answer the questions in the spaces provided. Your responses should include relevant mathematical reasoning and/or calculations. Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Question 5 (4 marks)

Marks

Lori lives with her friends in a rental property. Her annual salary is \$30 680.

Each week she pays \$100 as her contribution for rent, \$53 for transport, and \$42 for food.

She spends \$150 per week on entertainment, \$84 per fortnight on personal items, and \$380 per month on clothes, saving the rest of her income.

- a) Calculate each of Lori's **annual** expenses in the table below:

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Annual Income		Annual Expenses	
Salary	\$30 680	Rent	
		Transport	
		Food	
		Entertainment	
		Personal Items	
		Clothes	

- b) How much is Lori able to save in one year after paying for her expenses?

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- c) Lori wants to buy a car in 3 years using her savings. The cost of the car is \$18 000.

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Determine using calculations if Lori will be able to buy the car.

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Question 6 (4 marks)

Henry wants to purchase a new car valued at \$45 000. He will pay for the car and stamp duty with a loan.

Henry pays a \$5000 deposit for the car. The remaining amount, including the stamp duty, he borrows as a loan at a simple interest rate of 8% p.a., paid in equal monthly instalments over 3 years.

- a) Calculate the stamp duty on the car if stamp duty is charged at 3% of the vehicle price. 1

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- b) How much money does Henry borrow (including stamp duty)? 1

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- c) In total, how much does Henry need to repay? 1

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- d) Calculate each monthly instalment. Round to the nearest cent. 1

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Continue to Section III – Rates and Ratios

Section III – Rates and Ratios

8 marks

Attempt Questions 7–9

Allow about 10 minutes for this section.

Answer the questions in the spaces provided. Your responses should include relevant mathematical reasoning and/or calculations. Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Question 7 (3 marks)

Marks

A person's maximum heart rate (MHR) in beats per minute (bpm) is estimated using the formula:

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$$\text{MHR} = 220 - \text{age (years)}$$

Daniel is 20 years old and exercises at 70% of their maximum heart rate.

Daniel's heart pumps 70 mL of blood with each heartbeat.

Calculate the volume of blood pumped by his heart in 1 hour of exercise. Write your final answer in litres.

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Question 8 (2 marks)

Chloe uses a television with a rating of 60 W on average for 2 hours each day.

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How much does Chloe need to pay each year for electricity for using her television if electricity is charged at a rate of 30 cents per kWh. Answer to the nearest cent.

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Question 9 (3 marks)

Jessica drives her car to and from school 5 days a week. The distance between her home and school is 15 km. Her car has an average fuel consumption rate of 5.6 L/100 km.

- a) If petrol costs \$2.15 per litre, how much does Jessica have to pay each week for travelling to and from school?

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- b) Kittana travels to and from school 5 days a week by bus instead, paying \$1.64 for each bus trip.

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How much money does Kittana save compared to Jessica each week by travelling by bus, assuming Jessica only needs to pay for petrol?

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Continue to Section IV – Investments and Loans

Section IV – Investments and Loans

8 marks

Attempt Questions 10–12

Allow about 10 minutes for this section.

Answer the questions in the spaces provided. Your responses should include relevant mathematical reasoning and/or calculations. Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Question 10 (4 Marks)

Marks

Jasmine and Su wish to invest their savings into shares:

Together, they purchase 2500 shares of BHS which have a market price of \$1.85 per share.

The brokerage fee is set at 3% of the total value of the shares bought.

- a) What is the **total cost** of purchasing the shares, including the brokerage fee? 2

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- b) Jasmine and Su are paid a dividend of \$0.25 per share. Calculate the **total dividend**. 1

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- c) Hence, calculate the **dividend yield**. Round your answer to 1 decimal place. 1

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Question 11 (2 Marks)**Marks**

Giana invests \$15 000 over 5 years at an interest rate of 6% p.a. compounded monthly.

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Calculate the **interest** Giana earned at the end of 5 years, correct to the nearest cent.

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Question 12 (2 Marks)

Oscar purchased a computer which depreciates in value at a rate of 40% per annum.

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Three years after the purchase, it had depreciated to a value of \$540, using the declining balance method.

How much did Oscar **originally** pay for the computer?

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END OF EXAMINATION

If you use this space, clearly indicate which question you are answering.

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